

IoT-FKGD-L-SL: Anomaly Detection Framework Integrating Knowledge Distillation and a Swarm Learning for 5G IoT

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Abstract—Anomaly detection using multivariate time series (MTS) is critical for detecting abnormal traffic and device failures in 5G Internet of Things (IoT) devices. The current anomaly detection framework lacks the ability to model multidimensional long time series and to address issues, such as resource overhead, privacy protection, and data security in distributed learning modes within the IoT. Therefore, this article proposes an anomaly detection framework integrating knowledge distillation and swarm learning for 5G IoT (IoT-FKGD-L-SL). First, to model the correlations between different variables, a new method for capturing correlations between variables through clustering is proposed. Second, to perform long-term modeling of MTS, a long-time-series anomaly detection model called IoT-FKGD is proposed, based on multiscale dilated convolution and locality-sensitive hashing (LSH) attention. Finally, a framework based on IoT-FKGD is proposed to detect traffic anomalies of IoT devices under a swarm learning architecture that incorporates knowledge distillation. The effectiveness of the IoT-FKGD-L-SL framework is demonstrated by comparing it with advanced anomaly detection methods on real data sets. Experimental results show that on a long time scale, the precision, recall, and F1-score of anomaly detection using this framework all surpass those of baseline methods.

Index Terms—5G Internet of Things (IoT), anomaly detection, attention, graph convolution, knowledge distillation, swarm learning.

I. INTRODUCTION

5G INTERNET of Things (IoT) is an IoT application based on fifth-generation mobile communication technology, which can better meet the needs of IoT applications due to the 5G network's high-speed and low-latency, large-scale connectivity, and high-density connectivity, which is critical in realizing ultrahigh-speed, high-reliability, and low-latency data transmission [1]. Because of the deep integration of IoT technology and 5G, 5G IoT is widely employed in a variety of industries, including smart transportation, smart homes, smart medical care, smart factories, and so on. However,

when new technologies are introduced and 5G IoT scales up, new security vulnerabilities emerge. On the one hand, access to large-scale IoT devices might generate network congestion, resulting in anomalous device traffic. On the other side, IoT nodes may display anomalous behaviors that cause network disruptions, which can be caused by network assaults or device malfunctions [2]. Existing 5G IoT networks often include a significant number of sensor devices that generate traffic data in the form of time series. Therefore, through intelligent analysis of time-series traffic of 5G IoT devices, timely detection of abnormal traffic in the network is of great security significance for 5G IoT devices [3], [4], [5], [6].

Machine learning (ML) and deep learning approaches are now commonly utilized to extract temporal and spatial features from multivariate time series (MTS) for anomaly identification. Literature [7] presents an ensemble ML strategy based on mean shift and local outlier factor (LOF) for detecting anomalies in IoT devices. Literature [8] examines the cost-effectiveness of various ML algorithms, and the results show that gradient boosting machines, random forests, and deep learning models are the most accurate and fastest anomaly detection models. Gradient boosting machines and random forests are the most accurate and memory-efficient models, assisting the industry in selecting appropriate ML models for anomaly detection across various IoT networks. Literature [9] provides an IoT anomaly detection approach based on graph attention gated recurrent units (GAGRUs), which uses complicated spatiotemporal properties in MTS for modeling. Literature [10] presents a unique communication-efficient device-end federated learning (FL) deep anomaly detection system for sensing time-series data in the Industrial IoT (IIoT). Literature [11] introduces a model called BeatGAN, an adversarial learning reconstruction model that learns to detect anomalies based on normal data or data where the majority of the samples are normal, as infrequent anomalies can lead to severe reconstruction mistakes. Literature [12] describes an approach known as parallel attention, which differentiates anomalies by modeling MTS with global and local patterns. Literature [13] proposes an adversarial time-frequency reconstruction network (ATF-UAD) for unsupervised anomaly detection, which maximizes the reconstruction of normal values in sequences by combining time reconstruction, frequency reconstruction, and dual-view adversarial learning. All of these strategies have been shown to be effective and accurate at detecting anomalies in MTS.

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However, existing anomaly detection systems continue to suffer from the following difficulties.

- 1) When it comes to time-series modeling, the aforementioned papers ignore the potential interrelationships across variables in multivariate data. These potential associations reflect the correlation between IoT device traffic measurements that vary over time.
- 2) Research indicates that current IoT device anomaly detection technologies train and predict based on a short time frame, which prevents models from capturing and analyzing the long-term dependencies and dynamic changes of multivariate long time series. This limitation hinders a better understanding of the patterns behind normal data. Additionally, anomaly detection models do not adequately focus on the behavior evolution of IoT devices and performance evaluation over long time scales [14].
- 3) Furthermore, present centralized deep learning algorithms collect massive volumes of traffic data from multiple IoT devices and send it to a central server for training anomaly detection models. This learning method runs the risk of knowledge leaking and faces obstacles due to the variety of computational resources and data [15], [16].

To address the aforementioned issues, this article proposes an anomaly detection framework integrating knowledge distillation and swarm learning for 5G IoT (IoT-FKGDL-SL). The main work and innovations of this article are summarized as follows.

- 1) A novel method for capturing long-term dependencies among variables is proposed, which introduces fast dynamic time warping (FastDTW) as a distance metric instead of Euclidean distance, based on the time-series K -Means method. This allows highly correlated variables to be clustered into the same group. Then, within each cluster, feature graphs are constructed, and a graph convolutional network (GCN) is used to aggregate information from related variables, generating new time vector embeddings.
- 2) To achieve long-term modeling of MTS, a reconstruction model based on dilated convolution and locality-sensitive hashing (LSH) attention is proposed. This model extracts local correlations of the sequences by constructing multiscale dilated convolutions and then uses multiple hash functions to project similar sequence elements into hash buckets. Self-attention is performed within the hash buckets to capture long-term dependencies of multivariate sequences, and finally, the MTS is reconstructed through a multilayer perceptron (MLP) network.
- 3) An IoT device traffic anomaly detection framework under a swarm learning architecture incorporating knowledge distillation is designed. This framework addresses the issues of resource overhead, privacy protection, and data security in the centralized deep learning model of IoT by integrating knowledge distillation technology into the swarm learning architecture.

II. SYSTEM MODEL

A. 5G IoT Device Traffic Anomaly Detection Scenarios

In the IoT, there are U devices, and the time-series data of the i th device is $X_i = \{X_i^0, X_i^1, \dots, X_i^M\} \in \mathbb{R}^{N \times M}$, where $X_i^m = \{x_i^{m,1}, x_i^{m,2}, \dots, x_i^{m,N}\} \in \mathbb{R}^N$ represents the m th time-series variable. X_i comprises both the training data X_i^{train} and the test data X_i^{test} . The training data X_i^{train} contains unlabeled normal data, while the test data X_i^{test} contains a portion of the labeled anomalous data X_i^{lable} and the majority of the unlabeled normal data X_i^{unlable} . Given an observation window $W_t = \{X_i^{0,t-\tau+1}, \dots, X_i^{0,t-1}, X_i^{0,t}\}$ of length τ at time point t , the original sequence X_i can be converted into a time window sequence $W = \{W_1, W_2, \dots, W_M\}$ for use as training input. Traffic anomaly detection for IoT devices determines whether a device's MTS data is abnormal within the observation window. Calculate the anomaly score AS_t for W_t by comparing AS_t with a threshold to obtain anomaly results. This article divides IoT device traffic anomaly detection into two steps. First, construct the IoT-FKGDL anomaly detection model. Second, build an anomaly detection framework integrating knowledge distillation and swarm learning based on IoT-FKGDL.

B. Anomaly Detection Model IoT-FKGDL

The IoT-FKGDL model can model long-time correlations between different variables and reconstruct MTS. The model is primarily made up of a FastDTW-based time-series K -means clustering module, a GCN-based module for capturing long-term dependencies among variables, a domain information extraction module based on multiscale dilated convolution and a long-sequence reconstruction module based on LSH attention. The clustering module receives the time-series data X_i as input and returns the clustered set of time-series variables as output. The collection of clustered variables is supplied into the variable long-time correlation capture module, which captures the variables' long-term correlation by creating a feature graph G within each cluster and running GCN. To reconstruct lengthier time series, the embedding vectors created by GCN are first employed to retrieve the time-series' local neighborhood information by multiscale dilated convolution. Then, each row of the generated query matrix Q , key matrix K , and value matrix V is hashed separately, with similar query vectors, key vectors, and value vectors being assigned to the same bucket. Self-attention is performed within each hash bucket, and finally, the sequences' long-term dependencies are extracted through an MLP network.

C. Anomaly Detection Framework IoT-FKGDL-SL Integrating Knowledge Distillation and Collective Learning

Based on IoT-FKGD, this article designs an IoT-FKGDL-SL framework for IoT device traffic anomaly detection is created using a swarm learning architecture that integrates knowledge distillation. Deploy the IoT-FKGDL teacher model and the lightweight IoT-FKGD student model on edge

servers accompanying IoT devices to achieve efficient real-time anomaly detection. Edge servers collect traffic data from IoT devices and, after preprocessing, generate a local traffic data set XMN, where N denotes the number of batches and M represents the batch size. First, the IoT-FKGDL teacher model is trained using the local data set. Then, edge servers use the collective network to share the teacher model parameters. A virtual central server generates global teacher model parameters and distributes them to the edge servers. Each edge server uses the global teacher model parameters and local data set to guide the training of the lightweight IoT-FKGDL student model. The student model is responsible for real-time anomaly detection. SL allows edge servers to continuously collect new traffic data to supplement the local data set. SL periodically updates the global teacher model and local student model parameters to ensure the accuracy of anomaly detection.

III. IOT-FKGDL ANOMALY DETECTION MODELS

This section will go over the details of the IoT-FKGDL anomaly detection model proposed in this article, as shown in Fig. 1, which is made up of four key modules: 1) a FastDTW-based time-series K -means clustering module; 2) a GCN-based module for capturing long-term dependencies among variables; 3) a neighborhood correlation extraction using multiscale dilated convolution module; and 3) a long-term sequence reconstruction using LSH self-attention module. The aim of capturing long-term dependencies among variables is to obtain new embedded representations of these long-term dependencies. Multivariate long-sequence reconstruction uses multiscale dilated convolution to capture neighborhood correlations of long sequences and employs LSH attention to learn the latent features of these long sequences. Finally, reconstruction error is used to determine anomalies.

A. FastDTW-Based Time-Series K -Means Clustering

MTS typically consists of multiple time-series indicators, but not all indicators are correlated with each other. For example, in IoT device traffic anomaly monitoring, there is correlation between upload speed, download speed, and transmission delay, while sensor readings may not correlate with upload speed. The rate of change of sensor readings may correlate with the sensor readings themselves. Clustering variable indicators based on their correlations can help. Variables within the same cluster exhibit strong correlations, while those in different clusters show weak or no correlations. Capturing long-term dependencies within clusters for anomaly detection naturally enhances the accuracy of MTS reconstruction and thus improves anomaly detection accuracy overall.

In this work, considering the direct use of Euclidean distance as a similarity measure between time series cannot align nonaligned parts of the time series, making it difficult to capture their nonlinear relationships. However, FastDTW is more robust to noise and local deformations in time series and is more efficient than standard DTW when handling long time series. Therefore, this article utilizes FastDTW as the distance metric in time-series K -Means to cluster time-series variables into clusters. To begin, each time-series FastDTW distance

is calculated for any two time series $X = \{x_1, x_2, \dots, x_n\}$ and $Y = \{y_1, y_2, \dots, y_n\}$ in the MTS after downsampling the elements of the low-resolution time series X' and Y' specified as

$$X' = \left\{ \frac{x_{2i-1} + x_{2i}}{2} \middle| i = 1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\} \quad (1)$$

$$Y' = \left\{ \frac{y_{2i-1} + y_{2i}}{2} \middle| i = 1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\} \quad (2)$$

where $\lfloor \bullet \rfloor$ denotes rounding down. Determine a cumulative distance matrix D' for X' and Y' . Calculate the distances using the classical DTW formula

$$D'(i, j) = d(x'_i, y'_j) + \min\{D'(i-1, j), D'(i, j-1), D'(i-1, j-1)\} \quad (3)$$

where $d(x'_i, y'_j) = \|x'_i - y'_j\|$ represents the distance between two spots with limited resolution. Backtrack the optimal path P' on the cumulative distance matrix D' , beginning in the lower right corner of the $D'(n', n')$ and ending at $D'(0, 0)$, and choose the path that reduces the cumulative distance. The generation of path P' can be expressed in the following formulaic manner:

$$P' = \{(n', n')\} \cup \{(i', j') \mid \arg \min_{(i, j) \in N(i', j')} D'(i, j)\} \quad (4)$$

where $N(i', j')$ denotes the set of (i', j') 's neighboring positions. The local window is then defined using the optimal path P' extension

$$\text{window} = \bigcup_{(i, j) \in P'} \{(2i, 2j), (2i+1, 2j), (2i, 2j+1), (2i+1, 2j+1)\}. \quad (5)$$

Compute the cumulative distance matrix D between high-resolution time series X and Y within a local window

$$D(i, j) = d(x_i, y_j) + \min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\}. \quad (6)$$

The ultimate FastDTW distance between the two time series can be calculated using the cumulative distance matrix D 's lower right element

$$\text{distance} = D(n, n). \quad (7)$$

This distance represents the degree of resemblance between two time series; the smaller it is, the closer the two series are.

For an MTS data collection $X = \{X_1, X_2, \dots, X_M\}$, where X_i represents the i th time series. Randomly assign k clustering centers $C = \{C_1, C_2, \dots, C_k\}$, where C_i is the starting center. For each sequence X_i , compute its FastDTW distance from all centers C_j and assign X_i to the center C_j with the shortest distance

$$j_i = \arg \min_j \text{FastDTW}(X_i, C_j) \quad (8)$$

where $\text{FastDTW}(X_i, C_j)$ denotes the FastDTW distance between the time series X_i and the cluster center C_j . The new center C_j is then calculated by taking the average value of each cluster C_j in relation to the X_i of all members

$$C_j = \frac{1}{|S_j|} \sum_{X_i \in S_j} X_i \quad (9)$$

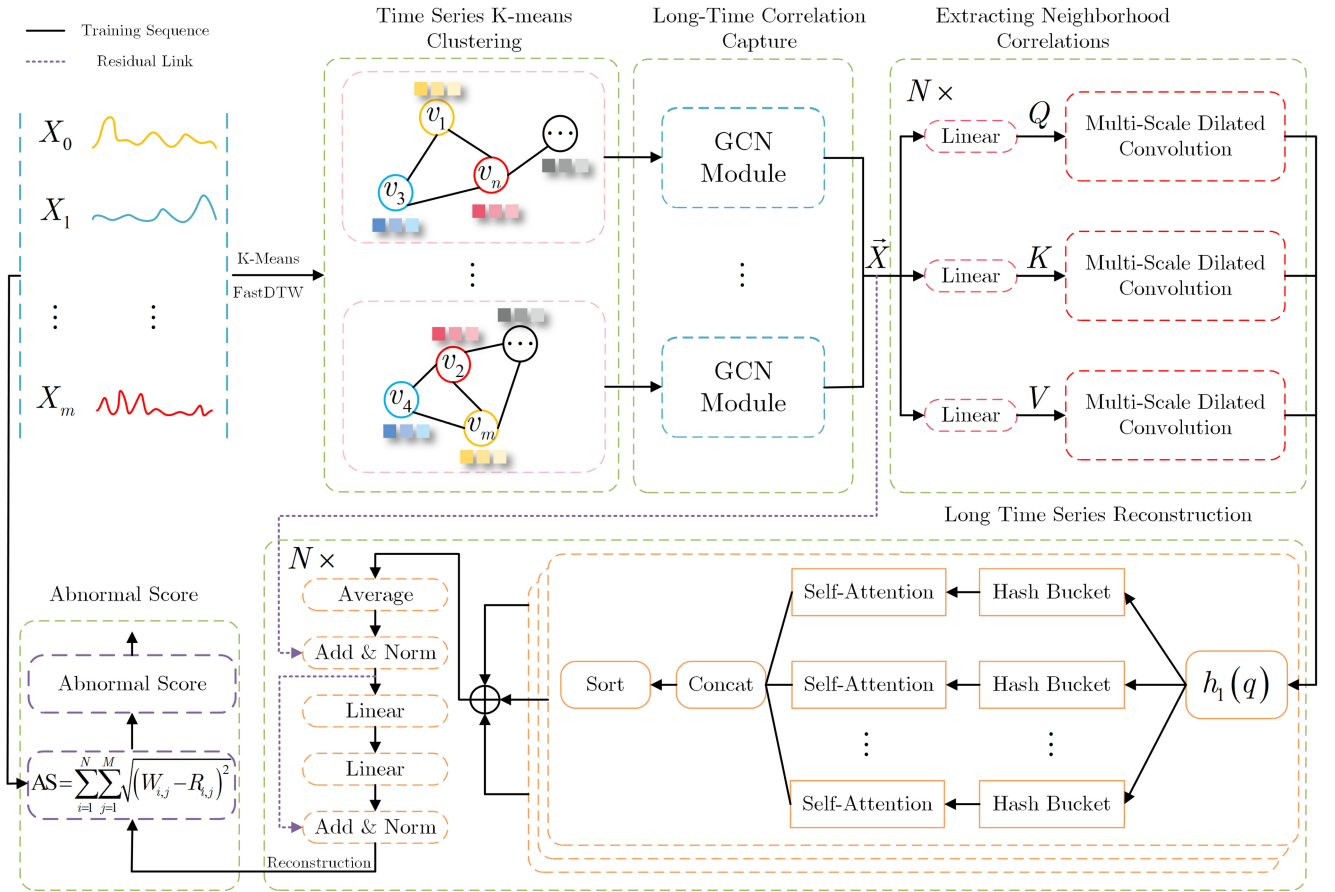


Fig. 1. IoT-FKGD anomaly detection models.

where S_j denotes the set of all time series assigned to cluster C_j . Finally, repeat (8) and (9) until the center stops changing.

B. GCN-Based Module for Capturing Long-Term Dependencies Among Variables

To better capture long-term correlation between variables, this work introduces the GCN model, which learns the correlation of variables inside each cluster. A feature graph $G_j = (V_j, E_j)$ is designed for time-series variables within a cluster C_j , with each node $v_i \in V_j$ representing a variable that is connected to the others, and the value of each variable in the current time window serving as its feature. The graph G_j is then subjected to a graph convolution procedure to combine variable features from surrounding nodes. According to [17], the following layer-by-layer propagation rule is used in multilayer convolutional networks:

$$H_\tau^{(l)} = \text{RELU}\left(\tilde{D}_\tau^{-\frac{1}{2}} \tilde{A}_\tau \tilde{D}_\tau^{-\frac{1}{2}} H_\tau^{(l-1)} W_\tau^{l-1}\right) \quad (10)$$

where $\tilde{A}_\tau = A_\tau + \lambda I_N$ is the adjacency matrix of time-series variables within cluster C_j , I_N is the unit matrix, and λ is the learnable parameter. Degree Matrix $\tilde{D}_\tau^{ii} = \sum_j A_\tau^{ij}$, $W_\tau^{l-1} \in \mathbb{R}^{m \times m}$ is the trainable weight matrix, and m is the layer's hidden feature dimension. The l th layer of the graph convolutional layer produces $H_\tau^l \in \mathbb{R}^{m \times m}$, while the nonlinear activation function is denoted by $\text{RELU}(\bullet)$.

Residual connections are introduced into the graph convolution layer to make the network's learned features more discriminative. The residual connections is represented as

$$H_\tau^{(l+1)} = \text{RELU}\left(\tilde{D}_\tau^{-\frac{1}{2}} \tilde{A}_\tau \tilde{D}_\tau^{-\frac{1}{2}} H_\tau^{(l)} W_\tau^l + H_\tau^l\right). \quad (11)$$

The output of graph convolution is the embedded vectors capturing correlations within cluster C_j , designated $\tilde{x}_j = H_j^{(l)}$.

C. Neighborhood Correlation Extraction Using Multiscale Dilated Convolution

The reconstruction module learns and interprets the underlying patterns in the data from the embedded vectors. By capturing and analyzing the long-term dependencies and dynamic changes of MTS, it provides a deeper understanding of the data characteristics. Normal data gives a very tiny reconstruction error, but abnormal data produces a huge reconstruction error since it does not follow the normal distribution. When dealing with longer MTS, conventional approaches frequently lose some local detail information. As a result, this work provides a reconstruction approach that uses multiscale dilated convolution and LSH attention to rebuild multivariate long time series.

The embedding vector $\tilde{X} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_N\} \in \mathbb{R}^{N \times M}$ of the MTS variables of the graph convolution output, $\tilde{x}_t \in \mathbb{R}^M$ represents an M -dimensional vector at time point t ($t \leq N$),

where N is the sequence length and M is the variable's dimensionality. First, the query, key, and value components are obtained using linear mapping as shown in

$$\begin{aligned} Q &= \tilde{X}W_Q + b_Q \\ K &= \tilde{X}W_K + b_K \\ V &= \tilde{X}W_V + b_V \end{aligned} \quad (12)$$

where $Q = \{q_1, q_2, \dots, q_M\} \in \mathbb{R}^{N \times M}$ signifies the query component, $K = \{k_1, k_2, \dots, k_M\} \in \mathbb{R}^{N \times M}$ denotes the key component, $V = \{v_1, v_2, \dots, v_M\} \in \mathbb{R}^{N \times M}$ denotes the value component, and q_i, k_i , and v_i represent the three matrices' column vectors. $W_Q \in \mathbb{R}^{m \times m}$, $W_K \in \mathbb{R}^{m \times m}$, and $W_V \in \mathbb{R}^{m \times m}$ symbolize the learnable weight vectors, while b_Q, b_K , and b_V represent the bias terms.

Due to the continuity of MTS generated by IoT devices, anomalies typically occur in localized regions where anomalous data points exhibit strong correlations with neighboring regions but weaker correlations with distant data points [18]. Expressing and utilizing these local relationships for anomaly detection naturally enhances the accuracy of anomaly detection in IoT device traffic. As a result, to construct the aggregated local information representation, 1-D extended convolution is conducted on q_i, k_i , and v_i , designated as

$$\begin{aligned} q'_i &= \text{DConv}(q_i; [f, r]) \\ k'_i &= \text{DConv}(k_i; [f, r]) \\ v'_i &= \text{DConv}(v_i; [f, r]) \end{aligned} \quad (13)$$

where $\text{DConv}(\bullet)$ symbolizes the dilated convolution function, $q'_i \in \mathbb{R}^N, k'_i \in \mathbb{R}^N, v'_i \in \mathbb{R}^N$, and N denote the sequence length, $1 \times f$ defines the size of the convolution kernel, and $r \in \mathbb{Z}^+$ represents sparsity. This indicates that at the moment t , f nearby time points can be aggregated. By default, we select sparsity $r = \{1, 2, 3\}$ and feel the wildness $f = \{3, 5, 7\}$.

D. Long-Term Sequence Reconstruction Using LSH Self-Attention

The self-attention mechanism has some drawbacks in modeling long sequences, most notably computational complexity and memory consumption. The computational complexity of the self-attention mechanism is $O(N^2)$, and as the length of the sequence increases, computation and memory consumption will cause an increase in the square level, and memory limitation will become the primary bottleneck in dealing with long sequences. In actual implementations, high computational and memory demands can create substantial delays, making real-time processing for IoT anomaly detection challenging to achieve. Furthermore, long-distance dependencies in large sequences are frequently sparse, with only a few parts having considerable dependencies. Due to the normalization procedure of attention weights, these long-distance dependencies may be diluted, preventing them from acquiring adequate attention weights.

Therefore, it is considered to partition the row vectors of the query matrix, key matrix, and value matrix of the long sequence into multiple buckets based on similarity. Self-attention is then performed within each bucket to effectively capture long-distance dependencies, thereby reducing

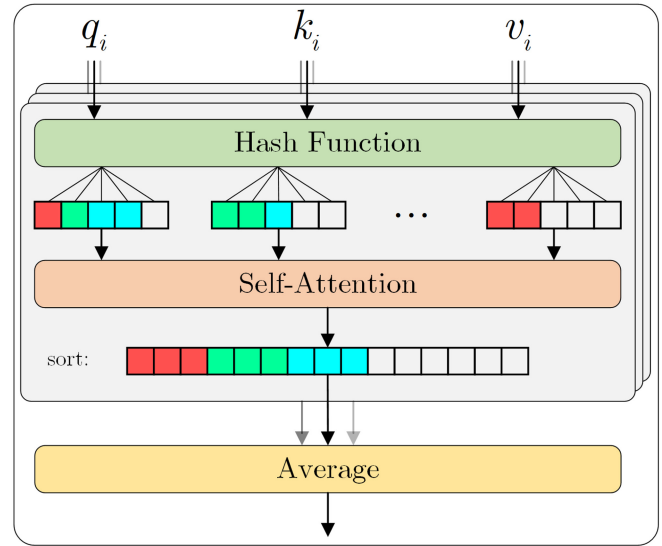


Fig. 2. Schematic of LSH attention.

the number of element pairs that need to be computed. Specifically, the computational complexity of LSH attention can be lowered to $O(N \log N)$, resulting in significant reductions in computational cost and memory consumption. The schematic of LSH attention is shown in Fig. 2. After extracting the neighborhood correlations of the time-series data, the query vector q_i , key vector k_i , and value vector v_i are mapped into hash buckets using multiple hash functions $h(q) = \{h_1(q), h_2(q), \dots, h_k(q)\}$. Using several hash algorithms increases the likelihood that comparable elements will be assigned to the same buckets, which is denoted as

$$h(q_i) = \left\lfloor \frac{q_i \cdot r_i + b_i}{\|r_i\|} \right\rfloor \quad (14)$$

$$h(k_i) = \left\lfloor \frac{k_i \cdot r_i + b_i}{\|r_i\|} \right\rfloor \quad (15)$$

$$h(v_i) = \left\lfloor \frac{v_i \cdot r_i + b_i}{\|r_i\|} \right\rfloor \quad (16)$$

where r_i is a random vector from the standard normal distribution and b_i is a random offset from the uniform distribution.

Because the amount of components in different buckets may be unevenly distributed, it is possible to have a bucket with a lot of q but relatively little k . Considering that buckets with nearby indices have higher element similarity, our strategy is to sort the obtained hash buckets by their indices, then sort the elements within each bucket according to their original order, and finally fix every c elements into a bucket sequentially. Self-attention is calculated within each bucket, marked as

$$\begin{aligned} \text{Attention}(Q_k, K_k, V_k) &= \text{softmax}\left(\frac{Q_k K_k^T}{\sqrt{M}}\right) V_k \\ 1 \leq k \leq \left\lfloor \frac{N}{c} \right\rfloor \end{aligned} \quad (17)$$

where M denotes the variable dimension. After getting the self-attention matrices for each bucket, they are stitched together to retrieve the initial positions

$$O_i = \text{sort}\left(\text{Concat}\left(\alpha_1, \alpha_2, \dots, \alpha_{\left\lfloor \frac{N}{c} \right\rfloor}\right)\right) \quad (18)$$

where $\alpha_i = \text{Attention}(Q_i, K_i, V_i)$, $\text{Concat}(\bullet)$ denotes matrix splicing, and $\text{Sort}(\bullet)$ denotes sorting. Following that, O_i created by different hash functions are combined and averaged to get the fused attention matrix, denoted as

$$O = \frac{1}{k} \cdot \sum_{i=1}^k O_i \quad (19)$$

where k denotes the number of hash functions. After obtaining the fused attention matrix O , residual linking and normalization is done by O and the embedding vector $\tilde{X}$ from Section III-B, and then the long time correlation R' in the sequence is captured more comprehensively by the MLP, and residual linking is done by R' with the normalization from the previous layer, which is finally reconstructed as R , which is denoted as

$$R' = \text{MLP}(\text{LayerNorm}(O + \tilde{X})) \quad (20)$$

$$R = \text{LayerNorm}(R' + \text{LayerNorm}(O + \tilde{X})) \quad (21)$$

where $\text{LayerNorm}(\bullet)$ denotes layer normalization, residual linking is intended to mitigate the problem of vanishing gradients and explosions, allowing the model to be trained at a deeper level.

E. Abnormal Score

In the anomaly detection scenario, the observation window is denoted as $W = \{w_1, w_2, \dots, w_m\} \in \mathbb{R}^{n \times m}$, where n is the sequence length and m is the variable dimension. The reconstruction sequence for this window is $R = \{r_1, r_2, \dots, r_m\} \in \mathbb{R}^{n \times m}$. In this article, the anomaly score AS is defined as the reconstruction error sum for each timestamp, denoted as

$$\text{AS} = \sum_{i=1}^n \sum_{j=1}^m \sqrt{(w_{i,j} - r_{i,j})^2}. \quad (22)$$

F. IoT-FKGD L Model Train

The reconstruction model is trained on the unlabeled normal data set X_i^{train} . We employ the mean-square error (MSE) between the reconstructed output $\hat{R} \in \mathbb{R}^{n \times m}$ and the observation R as a loss function, with the goal of decreasing

$$L_{\text{mse}} = \frac{1}{M} \sum_{t=1}^n \left\| \hat{R}^{(t)} - R^{(t)} \right\|_2^2. \quad (23)$$

After obtaining the error of the reconstructed model, the gradient of the reconstructed model is calculated by L_{mse} with the following formula:

$$\nabla \phi_{re} = \frac{1}{M} \sum_{m=1}^M \frac{\partial L_{\text{mse}}}{\partial \phi_{re}} \quad (24)$$

where $\nabla \phi_{re}$ represents the average gradient of M reconstructed model samples, and the parameters are updated using the Adam optimizer. Reconstruction model training is shown in Algorithm 1.

Algorithm 1 IoT-FKGD L Model Training Algorithm

Hyperparameters: Stacked Layers $layers$; Learning rate η ; Number of iterations T ; Batch size M ;

Input: Unlabeled normal dataset $\cup_{n=1}^N \{X_n^M\}$, where M denotes the batch size.

Output: The model convergence parameter W .

- 1: Randomly initialize k clustering centers $C = \{C_1, C_2, \dots, C_k\}$, where C_i is the starting center;
- 2: **repeat**
- 3: Eq. (7) calculates the FastDTW distance between each time series and all center C_i .
- 4: Using Eq. (8), allocate the time series to the nearest center C_j .
- 5: Eq. (9) calculates the average of all members as the new clustering center C .
- 6: **until** C No additional changes
- 7: **while** $t \leq T$ **do**
- 8: **for** $n = 1 : N$ **do**
- 9: Eq. (11) generates the embedding vector $\tilde{X}_n^M$ with correlation using X_n^M ;
- 10: **for** $l = 1 : layers$ **do**
- 11: Q' , K' , and V' are calculated from Eqs. (12)-(13) by extracting the neighborhood correlation using $\tilde{X}_n^M$;
- 12: From Eqs. (14)-(17), self-attention is determined from Q' , K' , and V' sub-buckets;
- 13: From Eqs. (18)-(19), fusing the self-attention of each barrel gives O_n^M ;
- 14: Eq. (20) generates $\tilde{R}_n^M$ based on $\tilde{X}_n^M$ and O_n^M , which further captures the long-term correlation;
- 15: Eq. (21) reconstructs the sequence from $\tilde{R}_n^M$ following residual concatenation and normalization to $\tilde{X}_n^M$;
- 16: **end for**
- 17: Eq. (23) calculates the model reconfiguration loss L_{mse} using X_n^M and $\tilde{X}_n^M$;
- 18: Eq. (24) calculates the gradient of model backpropagation using the L_{mse} $\nabla \phi_{re}$;
- 19: Update the model parameters based on the gradient $\nabla \phi_{re}$: $W \leftarrow W + \eta \cdot \nabla \phi_{re}$;
- 20: **end for**
- 21: **end while**
- 22: **return** The output model convergence parameter W .

IV. IOT-FKGD L ANOMALY DETECTION FRAMEWORK UNDER SWARM LEARNING ARCHITECTURE INCORPORATING KNOWLEDGE DISTILLATION

The IoT-FKGD L anomaly detection model proposed in Section III is suitable for a single IoT device. However, in the IoT, multiple devices work collaboratively, and the IoT-FKGD L model has many parameters, resulting in slower inference speed. Existing centralized deep learning approaches risk information leakage, in which private and sensitive raw data is sent to a centralized server and potentially abused by the server [19]. In this research, we introduce the SL architecture to solve the problems of information leakage and

data security in centralized deep learning. SL is an advanced decentralized FL architecture where data will only be trained locally and only model parameters are shared in the swarm network. This technique accomplishes an effective tradeoff between huge data and privacy [20]. However, the variety of users' computational capacity and data presents a barrier for SL designs as well. To overcome the difficulty, integrating knowledge distillation with SL design can effectively improve the model's knowledge and generalization capabilities while also reducing resource overhead. Knowledge distillation is a strategy for condensing deep learning models that involves taking the knowledge from the teacher model and transferring it to a smaller, lighter student model [21]. Fig. 3 depicts the workflow of this detection system, which includes data collection, local instructor model training, parameter sharing, local student model training, real-time anomaly detection, and continuous optimization. Given the limited processing capabilities of a single IoT device, edge servers representing many IoT devices are regarded as players in collaborative learning. Each participant should complete two steps during the training procedure. In the first step, each IoT device continuously monitors and gathers traffic data, which is preprocessed by the edge server before being used to train the teaching model locally with an anomaly detection method. The second stage is to upload the parameters of the local instructor model to the SL network for sharing with other participants. One edge server is chosen at random as the virtual central server, and it is in charge of aggregating the teacher model parameters submitted by each edge server to create a global teacher model. The global teacher model parameters are weighted based on the amount of node data or model performance to guarantee that the fused teacher model is more representative and resilient. The fused global teacher model is disseminated to each edge node, which combines the local data set and global teacher model to drive the training of lightweight student models. The trained student models are then utilized to detect traffic irregularities in IoT devices in real time. The trained student model is then used for real-time detection of traffic anomalies in IoT devices. SL enables edge servers to continually acquire new traffic data to enrich the local data set. Based on the new data, the edge servers train and update the local teacher model on a regular basis, and they share the most recent teacher model parameters across the SL network. Each edge server additionally concurrently updates the local student model, ensuring dynamic optimization of the entire student model. Compared to centralized data set-based training approaches, swarm learning architectures with knowledge distillation can guarantee data privacy while enabling collaborative learning and ongoing optimization. Furthermore, swarm learning can dynamically scale the number of edge nodes involved in learning, allowing it to adapt to the growing volume of IoT devices and data.

A. Teacher–Student Modeling Training

The edge servers of IoT devices all contain an IoT-FKGDL teacher model and an IoT-FKGDL student model, with the student model being considerably smaller and reasoning faster

than the teacher. Each edge server is in charge of gathering traffic data from local IoT devices and updating its local teacher and student models with each training cycle. Each edge server first trains the teacher model using the local data set (see Algorithm 1), after which the teacher model parameters are shared across the SL network. One edge server is chosen as the virtual center server, and the virtual center server receives the updated local teacher model parameters ϕ^{local} from each edge server. The global teacher model parameters are calculated using the weighted average algorithm [16] based on the amount of node data or model performance, which is aggregated as follows:

$$\phi^{\text{global}} = \sum_{s=1}^S \alpha_s \phi_s^{\text{local}} \quad (25)$$

where s denotes the s th edge server, and $\alpha_s \in [0, 1]$ denotes the weight of the aggregation.

Each edge server receives the global teacher model parameter ϕ^{global} published by the virtual center server, and the global teacher model guides the local student model training based on the local data set, which has the advantage of combining knowledge from other nodes and local knowledge to train the student model jointly. The distillation loss is defined as the L1 paradigm loss between the instructor and student model outputs, denoted by

$$L_{\text{KD}} = \sum_{i=1}^N \sum_{j=1}^M |X_{ij}^{(t)} - X_{ij}^{(s)}| \quad (26)$$

where $X^{(t)}$ denotes the output of the teacher model and $X^{(s)}$ denotes the output of the student model. The student model's total loss function is defined as the weighted sum of distillation and reconfiguration losses, denoted as

$$L = \alpha \cdot L_{\text{KD}} + \beta \cdot L_{\text{mse}} \quad (27)$$

where α, β represents the weight coefficient and L_{mse} is depicted in (28). After determining the loss of the student model, the gradient of the reconstructed model is computed from L using the following formula:

$$\begin{aligned} \nabla \phi^{(s)} &= \frac{\partial L}{\partial \phi^{(s)}} \\ &= \frac{\partial (\alpha \cdot L_{\text{KD}} + \beta \cdot L_{\text{mse}})}{\partial \phi^{(s)}} \\ &= \alpha \cdot \frac{\partial L_{\text{KD}}}{\partial \phi^{(s)}} + \beta \cdot \frac{\partial L_{\text{mse}}}{\partial \phi^{(s)}}. \end{aligned} \quad (28)$$

The training steps for the student model are shown in Algorithm 2.

Because IoT environments are dynamic, global models' detection accuracy may drop with time. As a result, constant optimization of this anomaly detection framework is necessary on an ongoing basis. Enabling the edge server to continuously collect and preprocess new traffic data by creating smart contracts to expand the local data set, and then continuously training and updating the local teacher and student models with the most recent data to improve the model's adaptability and accuracy. The virtual central server is responsible for

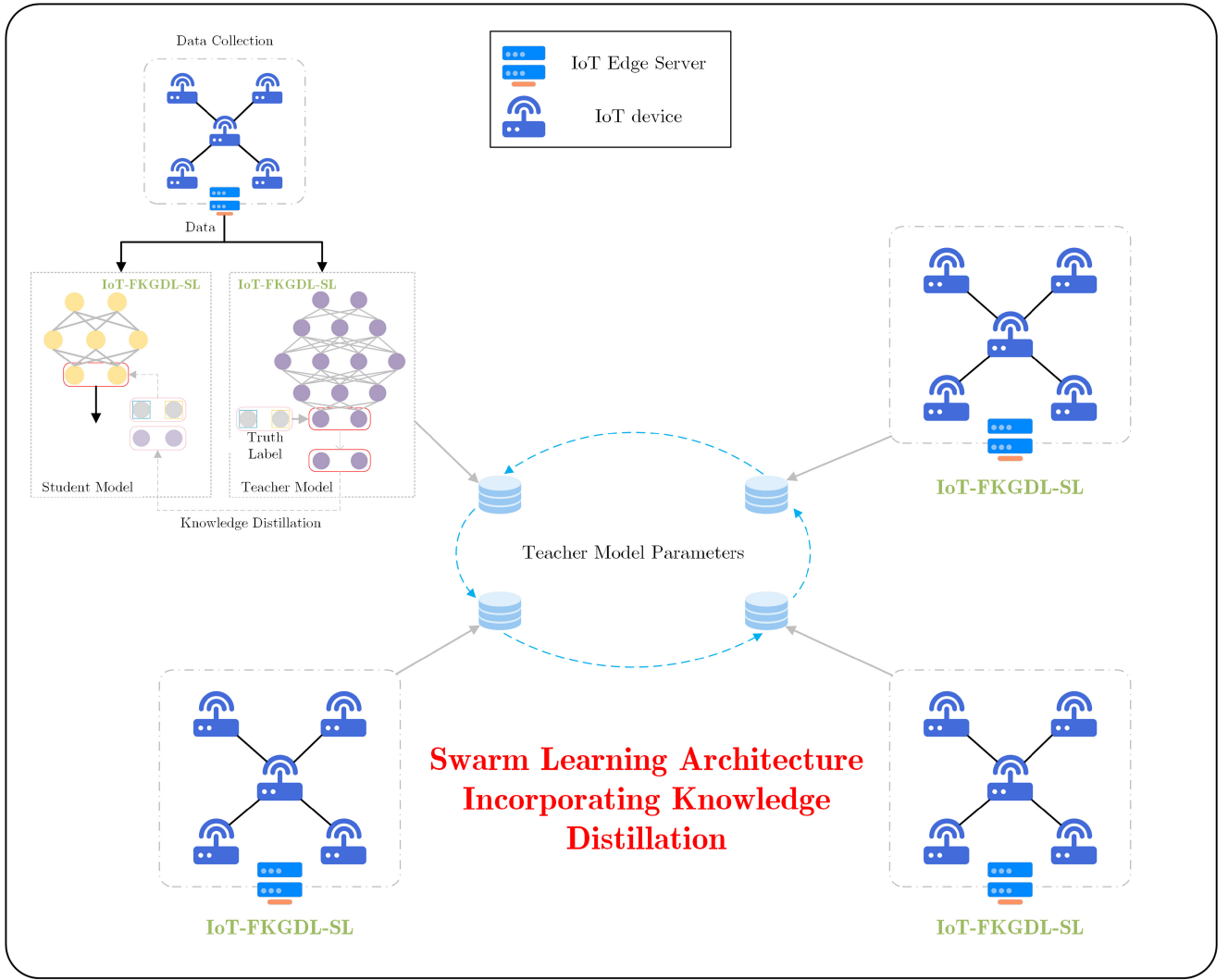


Fig. 3. IoT-FKGDL anomaly detection framework under a swarm learning architecture incorporating knowledge distillation.

periodically collecting and aggregating the latest teacher model parameters to create and disseminate a new global teacher model. This ensures the optimization and updating of the global teacher model, thereby enabling efficient anomaly detection and rapid response.

B. Real-Time Anomaly Detection

After student models have been trained for each edge server, they can detect anomalies in IoT device traffic in real time. An IoT device's anomalies may spread to surrounding devices as a result of its failure or network difficulties. If anomaly detection can be conducted in conjunction with surrounding devices' anomaly scores, the accuracy of anomaly detection will be considerably increased. Equation (22) enables the edge server to obtain the anomaly score AS_i of the local IoT device i . The average anomaly score of all nearby devices of device i is first calculated

$$\overline{AS}_k(i) = \frac{1}{k} \sum_{j=1}^k AS_j \quad (29)$$

where k denotes the k neighboring devices connected to device i . To integrate the anomaly scores of device i and its nearby devices, a weighted average approach is used

$$AS'_i = \alpha \cdot AS_i + (1 - \alpha) \cdot \overline{AS}_k(i) \quad (30)$$

where AS'_i is the composite anomaly score for device i , and $\alpha \in [0, 1]$ is a weighting factor used to balance the influence of one's own anomaly scores and those of one's neighbors. If the AS'_i exceeds the specified threshold ε , the system will generate an exception. This scoring technique integrates the anomalies of the device and its neighbors, preventing single-point anomaly scoring misjudgment, and allows the anomalies of IoT devices to be scored more precisely, boosting the accuracy and reliability of anomaly detection.

V. EXPERIMENTS

A. Experimental Setup

This experiment was run on a Linux server with 8 RTX3090 GPUs, using Python and Pytorch 2.0.1. The experimental data set, provided by the China Mobile Research Institute, is named

Algorithm 2 Training Algorithm for IoT-FKGDL Model Using Swarm Learning Architecture With Knowledge Distillation

Hyperparameters: Stacked Layers $layers$; Learning rate η ; Number of iterations T ; Batch size M ;

Input: Unlabeled normal dataset $\cup_{n=1}^N \{X_n^M\}$, where M denotes the batch size.

Output: Student model convergence parameters $W^{(s)}$.

- 1: Each edge server trains the teacher model parameters $W_i^{(t)}$ using Algorithm 1;
- 2: Eq. (25) aggregates the virtual central server with the global teacher model parameters $W^{(t)}$;
- 3: The edge server receives global teacher model parameters from the virtual center server $W^{(t)}$;
- 4: **while** $t \leq T$ **do**
- 5: **for** $n = 1 : N$ **do**
- 6: Calculate teacher model output $Z_n^{(t)}$ from X_n^M and teacher model parameters $W^{(t)}$;
- 7: Calculate student model output $Z_n^{(s)}$ from X_n^M and student model parameters $W^{(s)}$;
- 8: Distillation loss L_{KD} was calculated by combining $Z_n^{(t)}$ and $Z_n^{(s)}$ from Eq. (26);
- 9: The overall loss L of the student model is calculated according to Eq. (27);
- 10: The student model backpropagation gradient S is calculated from Eq. (28) based on the overall loss L ;
- 11: Update the student model parameters using the gradient $\nabla\phi^{(s)}$: $W \leftarrow W + \eta \cdot \nabla\phi^{(s)}$;
- 12: **end for**
- 13: **end while**
- 14: Algorithm 2 is executed on a periodic basis to dynamically optimize the teacher model parameter $W_i^{(t)}$ and the student model parameter $W_i^{(s)}$ for each edge server.

TABLE I
FEATURES OF THE LW5G-KPI DATA SET USED FOR MODEL TRAINING

Data set characteristics
Number of bytes received by the network interface in 2 minutes
Number of packets received by the network port in 2 minutes
Average rate received at the network interface over a 2-minute period
Average rate of data received at the network interface over a 2-minute period
Number of bytes transferred from the network port in 2 minutes
Number of packets transmitted through the network interface in 2 minutes
Average rate of network port transmissions over 2 minutes
Average rate of data transmitted through the network interface over a 2-minute period

LW5G-KPI. After data processing and feature engineering, eight indicators were selected from the data set to validate the model's performance. The features of the data set are shown in Table I. In this research, 70% of the data set is used for training, with the remaining 30% for validation. Adam is used as the optimizer during training, with a learning rate of 0.0001, a batch size of 32, and an anomaly detection window size of 8.

In this experiment, BeatGAN [11], SLA-VAE [22], MTAD-GAT [23], MAD-GAN [24], and LSTM-VAE [25] were selected as comparison models. On the LW5G-KPI data set,

we conducted comparative experiments on reconstruction loss and anomaly detection performance with the proposed model and comparison models, setting the time-series lengths to 24, 48, 96, 192, 384, and 576. Then, we conducted ablation experiments on the proposed model to validate the contribution of each component to the model's anomaly detection performance. Finally, we evaluated the impact of varying the number of participants in the group learning, set to 1, 10, 20, 50, and 100, on the anomaly detection performance of IoT-FKGDL-SL.

B. Evaluation Metrics

We chose a collection of assessment measures that have been commonly utilized in trials to assess the IoT-FKGDL model. MAE, MSE, and RMSE are employed as evaluation metrics for reconstruction of lengthy time series, defined as follows:

$$\text{MAE}(y, \hat{y}) = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i| \quad (31)$$

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 \quad (32)$$

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2} \quad (33)$$

where y is the original sequence, $\hat{y}$ is the reconstructed sequence, and m represents the sample size.

To evaluate the model's anomaly detection ability, we use the most common precision rate, recall rate, and F1 score as model evaluation metrics, as specified in the following equations:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (34)$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (35)$$

$$\text{F1} = 2 \times \frac{\text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \quad (36)$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively.

C. Performance Analysis of Multivariate Long Time-Series Reconstruction

In Fig. 4, we compare the reconstruction loss of our proposed IoT-FKGDL to other models on the LW5G-KPI data set with input sequence lengths of 24, 48, 96, 192, 384 and 576, respectively. It can be shown that for sequence lengths of 24 and 48, the reconstruction losses MAE, MSE, and RMSE of each model are reasonably steady and minimal, indicating that the models can recreate the data more accurately at smaller time scales. When the sequence length reaches 96, the other models' reconstruction losses begin to grow, while the IoT-FKGDL remains rather steady. The increasing trend of the MAE, MSE, and RMSE values of the BeatGAN, SLA-VAE, MTAD-GAT, MAD-GAN, and LSTM-VAE models is more

significant when the sequence length increases to 192, 384, and 576. This suggests that these models can no longer accurately reconstruct the data as the sequence length increases, which will have an impact on the anomaly detection results. Our suggested IoT-FKGDL has reconstruction losses that are all much smaller than those of alternative techniques, and its accuracy is within acceptable bounds. Additionally, the values of MAE, MSE, and RMSE stay comparatively constant as sequence length increases. IoT-FKGDL performs better for longer MTS reconstruction than other models, according to the experimental data.

D. Anomaly Detection Performance Analysis

In Table II, we compare the anomaly detection performance of our proposed IoT-FKGDL to other approaches for input sequence lengths of 24, 48, 96, 192, 384, and 576. On the LW5G-KPI data set, IoT-FKGDL achieves the most precision, BeatGAN achieves the highest recall, and SLA-VAE achieves the highest F1 score when the model reconstruction sequence is 24. When the sequence length is 48, IoT-FKGDL outperforms other anomaly detection algorithms by just 0.00–0.019, 0.007–0.016, and 0.007–0.014 in precision, recall, and F1 score, respectively. This shows that the anomaly detection performance of each method is more satisfactory at lower time scales. However, when the sequence length is 192, IoT-FKGDL outperforms other anomaly detection methods by 0.073–0.089 in accuracy, 0.025–0.031 in recall, and 0.023–0.061 in F1 score. When the sequence length is 384, IoT-FKGDL outperforms other anomaly detection methods by 0.024–0.146 in accuracy, 0.033–0.077 in recall, and 0.041–0.082 in F1 score. When the sequence length is 576, IoT-FKGDL outperforms other anomaly detection methods by 0.114–0.201 in accuracy, 0.076–0.128 in recall, and 0.078–0.130 in F1 score. Table II shows that the precision, recall, and F1 scores of the other models steadily drop as the sequence length rises, whereas the evaluation metrics of IoT-FKGDL remain reasonably stable or slightly increase.

Stability of anomaly detection is also critical in IoT device traffic anomaly detection. We run ten tests on each model to assess the stability of the algorithms for anomaly identification over extended input sequence lengths using important evaluation metrics, such as precision, recall, and F1 score. Fig. 5 depicts the experimental results, which show that the IoT-FKGDL approach has the highest stability in extended time scales, indicating that it can retain constant detection performance across long time scales. BeatGAN, SLA-VAE, and MTAD-GAT algorithms have somewhat worse precision, recall, and F1 scores than the IoT-FKGDL approach. While the MAD-GAN and LSTM-VAE algorithms shown significant performance swings throughout testing, this instability may limit their reliability in anomaly detection over extended time scales. Detecting anomalies over long time periods is especially difficult in IoT environments, and the IoT-FKGDL technique is better suited to the task. Experimental results demonstrate that the IoT-FKGDL technique is adaptable and robust in detecting IoT device traffic anomalies over long time periods.

E. Ablation Experiment

We divided the components into four categories: 1) *K*-means clustering; 2) GCN; 3) multiscale dilate convolution; and 4) LSH, and tested their performance for anomaly identification. For the discriminative criterion, we examine the impact of utilizing a fixed threshold versus an SL-threshold that aggregates the anomaly scores of surrounding nodes on detection performance. Table III shows that each component contributes to the final anomaly detection, with the first and sixth sets of tests improving precision, recall, and F1 scores by 0.053, 0.006, and 0.014, respectively, for multiscale dilate convolution. The second set of studies used the self-attention mechanism rather than LSH attention to confirm the role of LSH attention in detection performance. It can be observed that using LSH attention improves the model by 0.052 in precision, 0.001 in recall, and 0.010 in F1 score. The third and fourth sets of trials demonstrated the validity of utilizing GCN to capture long-term correlations between variables. A comparison of the fourth and sixth sets of tests confirms the efficacy of employing *K*-means clustering and recording variable correlations within clusters to find model anomalies. These two sets of experiments indicate that our proposed method of first clustering variables and then capturing the correlations between them can improve the model by 0.039 in precision, 0.011 in recall, and 0.015 in F1 score. The fifth and sixth sets of experiments demonstrate that SL-Threshold considerably enhances the model's correct anomaly identification. It can be observed that using the SL-Threshold method improves the model by 0.023 in precision, 0.013 in recall, and 0.014 in F1 score. This is owing to the fact that SL-Threshold aggregates the abnormalities of surrounding nodes, increasing confidence in the identification of IoT devices.

F. Impact of the Swarm Learning Architecture Incorporating Knowledge Distillation

Table IV compares the time and space complexity of the IoT-FKGDL teacher and student models, as well as providing model specifics. The teacher and student models have the same theoretical time and space complexity of $O(L \log(L))$. We set the number of hidden neurons in the GCN layer for the teacher and student models to 2048 and 512, respectively, as well as the number of hidden neurons in the MLP layer to 2048 and 512. We also categorize the hash functions as 8 and 4. The model's size was lowered while retaining the performance of the student model. Fig. 6 compares the total runtime of the IoT-FKGDL teacher model and the student model during the training and testing phases, revealing that the IoT-FKGDL-Student has the maximum training efficiency. During the testing phase, IoT-FKGDL-Student outperformed IoT-FKGDL-Teacher. This demonstrates how knowledge distillation can improve the pace of model inference.

We set the number of participants in the swarm learning network to $S \in \{1, 10, 20, 50, 100\}$, and we use the IoT-FKGDL-SL student model for anomaly detection, obtaining precision, recall, and F1 score as model evaluation metrics and comparing them to the independently trained IoT-FKGDL-SL student model ($S = 1$). As shown in Figs. 7–9, when the

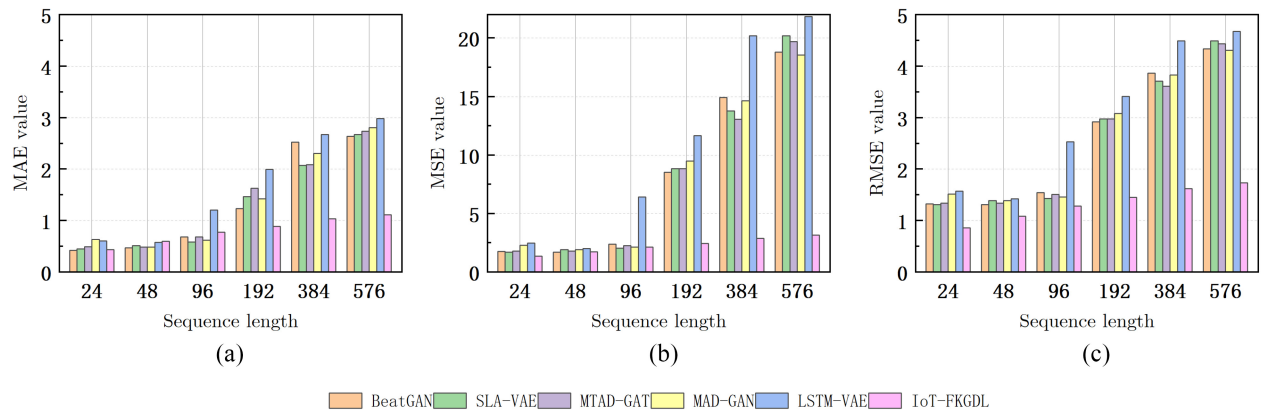


Fig. 4. Comparative analysis of reconstruction losses for each model with different sequence lengths. (a) MAE values vary with sequence length. (b) MSE values vary with sequence length. (c) RMSE values vary with sequence length.

TABLE II
IMPACT OF EACH ANOMALY DETECTION AT VARIOUS SEQUENCE LENGTHS ON THE LW5G-KPI DATA SET, WITH LARGER VALUES INDICATING GREATER PERFORMANCE

Methods	IoT-FKGDL			BeatGAN			SLA-VAE			MTAD-GAT			MAD-GAN			LSTM-VAE		
Metrics	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
24	0.968	0.882	0.910	0.965	0.906	0.927	0.964	0.903	0.925	0.953	0.897	0.918	0.959	0.905	0.925	0.948	0.901	0.920
48	0.974	0.880	0.909	0.962	0.873	0.902	0.974	0.868	0.899	0.967	0.864	0.896	0.962	0.871	0.900	0.955	0.865	0.895
96	0.975	0.882	0.911	0.951	0.879	0.905	0.947	0.893	0.915	0.936	0.892	0.912	0.927	0.869	0.0.893	0.917	0.872	0.894
192	0.971	0.916	0.936	0.947	0.891	0.913	0.935	0.884	0.906	0.927	0.871	0.895	0.916	0.870	0.892	0.882	0.855	0.875
384	0.976	0.905	0.928	0.903	0.858	0.881	0.886	0.870	0.887	0.884	0.862	0.881	0.854	0.872	0.882	0.830	0.828	0.846
576	0.982	0.918	0.939	0.858	0.842	0.861	0.868	0.827	0.851	0.837	0.822	0.842	0.819	0.833	0.847	0.781	0.790	0.809

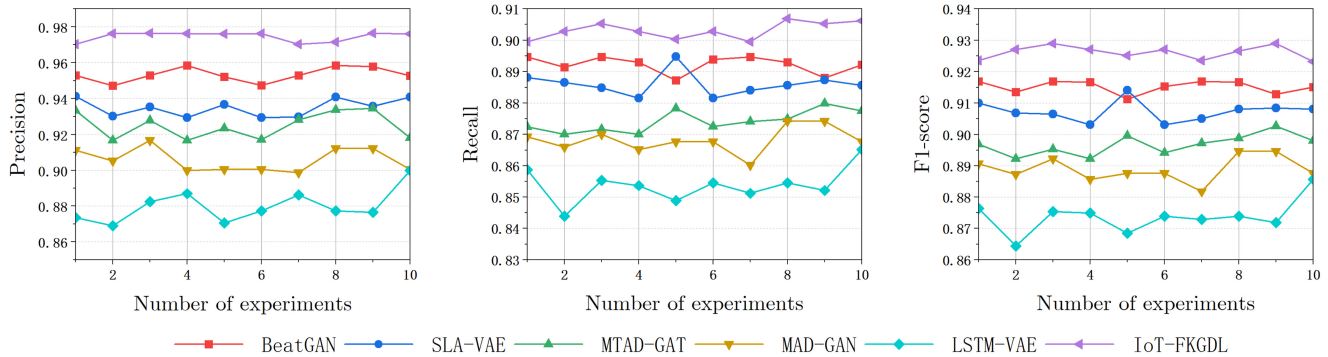


Fig. 5. Stability examination of precision, recall, and F1 score metrics for each model trained ten times separately.

TABLE III
ABLATION OUTCOMES AND ANOMALY CRITERIA FOR COMPONENTS IN IOT-FKGDL

Components			Exception Standards			Evaluation metrics		
K-means clustering	GCN	Multi-scale Dilate Convolution	LSH	Thresholds	SL-Threshold	Precious	Recall	F1-score
✓	✓	✗	✓	✗	✓	0.929	0.912	0.925
✓	✓	✓	✗	✗	✓	0.930	0.917	0.929
✗	✗	✓	✓	✗	✓	0.943	0.907	0.924
✗	✓	✓	✓	✗	✓	0.969	0.892	0.917
✓	✓	✓	✓	✓	✗	0.959	0.905	0.925
✓	✓	✓	✓	✗	✓	0.982	0.918	0.939

number of participants ranges from 1 to 20, precision, recall, and F1 score remain relatively stable. However, as the number of participants increases to 100, the performance of anomaly detection gradually decreases, with precision dropping from a maximum of 0.984–0.943. Meanwhile, recall and F1 score

remain relatively stable, with recall decreasing by only 0.009 and F1 score decreasing by only 0.0116. This experiment validates the effectiveness of IoT-FKGDL-SL with knowledge distillation in anomaly detection within a swarm learning framework.

TABLE IV
COMPLEXITY ANALYSIS AND DETAILED PRESENTATION OF IoT-FKGDL TEACHER AND STUDENT MODELS

Methods	Time	Memory	Number of GCN hidden neurons	Number of MLP neurons	Number of hash functions
IoT-FKGDL-Teacher	$O(L \log(L))$	$O(L \log(L))$	2048	2048	8
IoT-FKGDL-Student	$O(L \log(L))$	$O(L \log(L))$	512	512	4

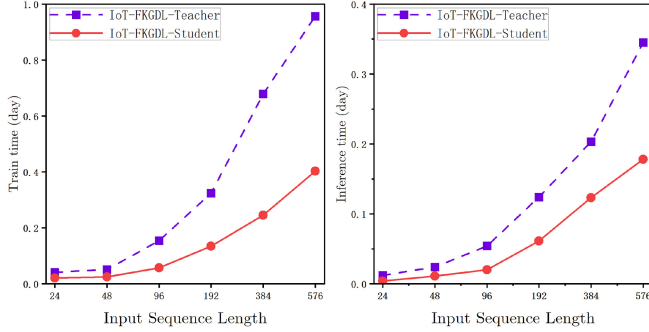


Fig. 6. Total runtime of training-testing phase.

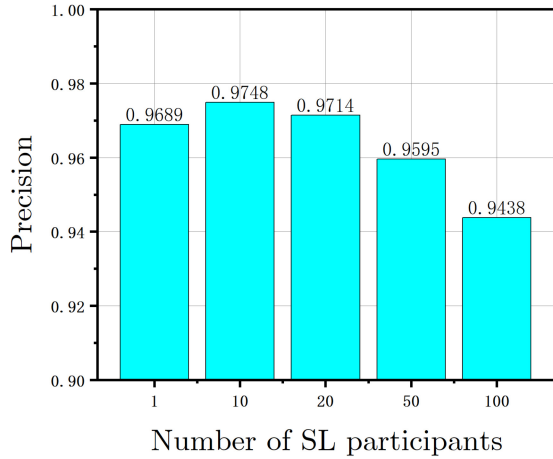


Fig. 7. Precision of model anomaly detection in different scenarios.

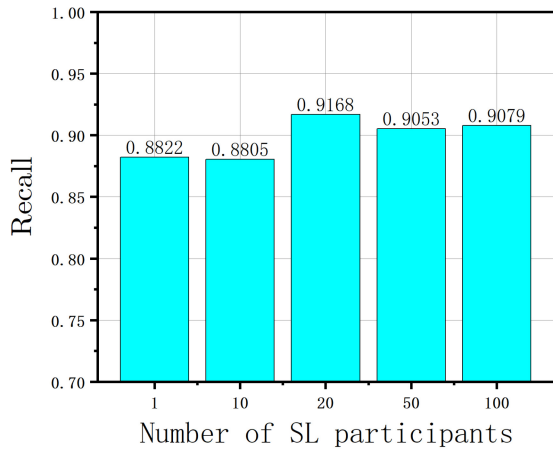


Fig. 8. Recall of model anomaly detection in different scenarios.

VI. CONCLUSION

Due to the critical importance of traffic anomaly detection for the normal operation of IoT systems, this article proposes a 5G IoT anomaly detection framework, IoT-FKGDL-SL, which integrates knowledge distillation and swarm learning.

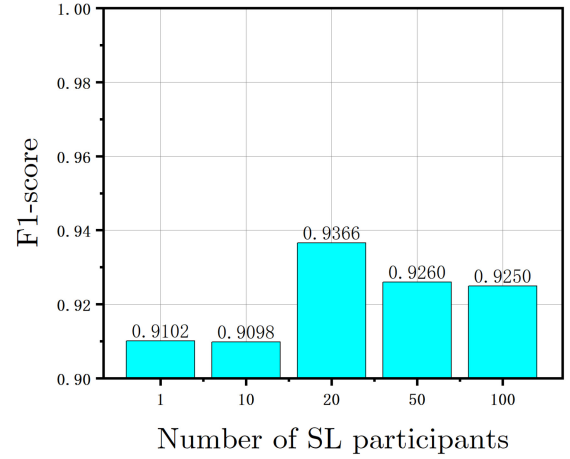


Fig. 9. F1 score for model anomaly detection in different scenarios.

The framework aims to enhance the modeling capability of multivariate long-term time series and address issues, such as resource overhead, privacy protection, and data security in distributed learning models within the IoT. We propose a novel method for modeling long-term correlations among time-series variables. By introducing FastDTW as the distance metric between time-series variables and combining it with time-series K -Means clustering for highly correlated variables, we effectively capture the long-term correlations between variables within each cluster using GCNs. Furthermore, we propose a long-time-series anomaly detection model, IoT-FKGDL, which employs multiscale dilated convolution to extract local correlations from the sequence. It combines LSH attention to capture the long-term correlations of multivariate sequences and utilizes an MLP network for the reconstruction of MTS. This article also designs an IoT-FKGDL-SL anomaly detection framework based on IoT-FKGDL, incorporating knowledge distillation within a swarm learning architecture to establish a global anomaly detection framework for IoT environments. Experimental results demonstrate that the IoT-FKGDL model can effectively reconstruct MTS over long time scales and outperforms baseline algorithms in precision, recall, and F1-score for anomaly detection. This validates the effectiveness and stability of our anomaly detection framework in handling IoT device traffic anomalies. Previous research has largely overlooked the model's capability for anomaly detection in multivariate long-term time series. Our study not only provides a new anomaly detection framework but also showcases its strong modeling and anomaly detection capabilities for multivariate long-term time-series data in practical applications. Particularly in scenarios requiring real-time response and distributed computing, the design of the IoT-FKGDL-SL framework effectively addresses issues related to insufficient computational resources and data privacy concerns faced by existing centralized methods.

The framework also has the following issues that need improvement. First, the experimental data sets in this study are primarily focused on specific types of 5G IoT devices, which may not cover all possible device types and anomaly patterns. The limitations of such data sets may affect the model's generalization ability, making it perform poorly with unseen device types or anomaly patterns. Our next steps involve considering more diverse data sets or conducting cross-device and cross-scenario testing to validate the framework's universality. Additionally, although the IoT-FKGDL-SL framework employs knowledge distillation for model compression, the computational complexity remains high when dealing with large-scale multidimensional time-series data. Future research could explore more efficient computing methods or model compression techniques to reduce the computational burden. Finally, while the model has shown significant improvements in anomaly detection performance, interpreting the model's decision-making process and anomaly detection results remains an important issue. Future work could focus on enhancing the model's interpretability to provide not only superior detection results but also clear explanations and justifications.

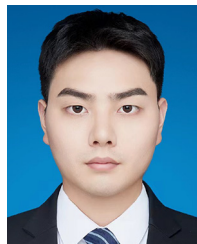
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